



Maveb Build Tool

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1. What is build tool

- A build tool is a software program that automates the process of transforming source code into a usable or executable program.
- Basic tasks
 - Compilation of source code
 - Linking required libraries
 - Testing
 - Packaging for easy distribution and deployment
 - Documentation etc
- Benefits of using build tools
 - Saves developers time and effort
 - Improved Consistency
 - Better Collaboration
 - Streamlined Development Process

- Maven requires Java to be installed on your system.
- Installation Steps for Windows
 - Download the binary archive
 - Extract the archive
 - Set environment variables
 - Verifying the Installation
- Eclipse with maven support
 - Eclipse offers built-in support for Maven through the M2Eclipse plugin
 - Eclipse allows to create new maven projects or allows to import existing maven projects.
 - Dedicated editors for the pom.xml file
 - Allows to execute Maven goals (like clean, install, package) from within Eclipse

- Project Object Model (POM):
 - Maven project is based on pom.xml file which defines various aspects of the project like
 - Project information
 - Dependencies
 - Build configuration
- Maven Repositories
 - Maven retrieves dependencies from repositories which store jar files.
 - The default repository is Maven Central. Any other custom repository can be set.
- Build Lifecycle
 - The build process is split up into build life cycles.
 - A build life cycle consists of a sequence of build phases, and each build phase consists of a sequence of goals.

- Some common phases include : clean, validate, compile, test, package, install, deploy
- Maven has 3 built-in build life cycles: default, clean and site
- Plugins
 - A set of actions for the project which are not covered by the standard Maven build phases and goals
- Profiles
 - Profiles allow to configure build settings for various environments like development or production
- Dependency Management
 - Maven automatically downloads the required libraries and their transitive dependencies from repositories like local repository, central repository and remote repository in the same sequence

- Maven has a standard directory structure

```
myprojectdir
- pom.xml

- .mvn
  - jvm.config

- src
  - main
    - java
    - resources
  - test
    - java
    - resources

- target
```